TASK 8 REPORT: BINARY MORPHOLOGICAL PROCESSING - EROSION, DILATION & ITERATIVE SHAPE MODIFICATION

- **Project Title:** Efficient Visual Token Pruning for Vision-Language Question Answering: A Comparative Study of Saliency-Based Reduction in LLaVA
- **Class / Branch:** EXCS-B
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1. Aim of the Experiment

To understand, implement and quantitatively evaluate fundamental binary morphological operations by:

1. Converting pre-processed and segmented chart representations into inverted binary foreground masks across multiple threshold values.
2. Implementing binary **Erosion** ($A \ominus B$) using 3×3 Square, 3×3 Cross-shaped and 5×5 Square Structuring Elements (SE) to observe structural shrinkage, noise removal and gap widening.
3. Implementing binary **Dilation** ($A \oplus B$) across the same structuring elements to evaluate object expansion, internal pinhole filling and component bridging.
4. Implementing Morphological Boundary Extraction ($\beta(A) = A - (A \ominus B)$) and analyzing morphological duality.
5. Evaluating multi-iteration erosion and dilation sequences ($k = 1,2,3$) to track the progressive elimination of text characters and the merging of adjacent chart data bars.
6. Mapping morphological filtering to patch-level spatial mask generation for Vision Transformer (ViT) visual token pruning in LLaVA-1.5.

2. Task 8.1: Binarization & Threshold Sensitivity Analysis

The input image was initialized from the pre-processed output established in prior modules (Median-denoised grayscale matrix of chart_01.png, resolution 600×850). Binarization was applied using inverted thresholding (cv2.THRESH_BINARY_INV) such that graphical data marks (bars, axis lines and text) map to foreground white (255), while background whitespace maps to black (0):

$$g(x,y) = \begin{cases} 255 & \text{if } f(x,y) < T \\ 0 & \text{otherwise} \end{cases}$$

Quantitative Threshold Comparison:

Threshold Mode	Threshold Value (T)	Foreground Area (Pixels)	Canvas Ratio (%)	Visual & Structural Assessment
Strict Cut-Off	$T = 150$	180,865	35.46%	**Under-segmentation:** Erases light pastel-shaded bars and thin font strokes.
Otsu's Automatic	$T = 185.0$	187,312	**36.73%**	**Optimal Baseline:** Maximizes inter-class variance; captures all bars and text.
Permissive Cut-Off	$T = 220$	196,265	38.48%	**Over-segmentation:** Captures faint compression halos and gridlines.



Figure 1: Thresholding evaluation on chart_01.png: Original RGB (far-left), Strict Threshold $T = 150$ (center-left), Otsu's Optimal Threshold $T = 185.0$ (center-right) and Permissive Threshold $T = 220$ (far-right).

3. Structuring Elements Definition

Binary morphology tests the local geometry of an image set A using a structuring element B:

1. 3×3 **Square SE (B_{rect3}):** A 3×3 matrix of all ones. Applies isotropic 8-neighbor morphological filtering.
2. 3×3 **Cross-Shaped SE (B_{cross}):** A cross kernel with 1s on the cardinal axes and 0s on the diagonals. Applies 4-neighbor morphological filtering.
3. 5×5 **Square SE (B_{rect5}):** A 5×5 matrix of all ones. Simulates larger two-pixel-radius neighborhood transformations.

4. Tasks 8.2 & 8.3: Implementation of Erosion and Dilation

A. Binary Erosion ($A \ominus B$)

Erosion tests whether the structuring element B translated to location z is fully contained within the foreground set A:

$$A \ominus B = \{z \in \mathbb{Z}^2 \mid (B)_z \subseteq A\}$$

Effect: Shrinks foreground objects, strips isolated boundary pixels, widens background gaps and eliminates thin line features smaller than the structuring element.

B. Binary Dilation ($A \oplus B$)

Dilation tests whether the reflected structuring element $\hat{B}$ translated to location z overlaps the foreground set A by at least one pixel:

$$A \oplus B = \left\{ z \in \mathbb{Z}^2 \mid \left(\hat{B}\right)_z \cap A \neq \emptyset \right\}$$

Effect: Expands foreground objects, bridges broken text strokes, fills internal pinholes and closes narrow gaps between neighboring components.

Quantitative Impact Across Structuring Elements:

Morphological Operation	Structuring Element	Foreground Pixels	Area Change (%)	Structural Impact on Chart Elements
Input Baseline	None (Otsu Binary)	187,312	0.00%	Full unmanipulated segmented foreground.
Erosion	3 × 3 Square	165,962	**-11.40%**	Uniform boundary shrinkage; strips single-pixel scan noise.
Erosion	3 × 3 Cross	166,975	**-10.86%**	Directional contraction; preserves diagonal font connectivity.
Erosion	5 × 5 Square	150,828	**-19.48%**	Severe contraction; disconnects text labels from bar axes.
Dilation	3 × 3 Square	213,771	**+14.13%**	Closes character loops; solidifies data bar cores.
Dilation	3 × 3 Cross	209,861	**+12.04%**	Orthogonal expansion; preserves diagonal baseline gaps.
Dilation	5 × 5 Square	240,257	**+28.27%**	Aggressive expansion; merges closely spaced parallel bars.

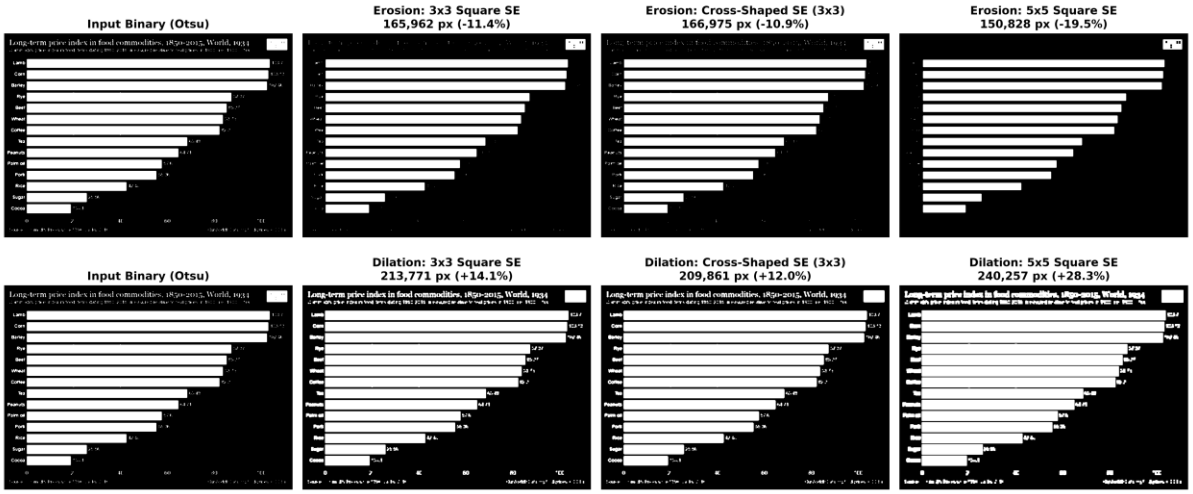


Figure 2: Structuring element comparison: Top row shows Erosion across 3 × 3 *Square,* 3 × 3 *Cross and* 5 × 5 *Square kernels; Bottom row shows Dilation across the identical structuring elements.*

5. Task 8.4: Comparative Analysis & Morphological Boundary Detection

Erosion and dilation exhibit mathematical **duality** with respect to set complementation:

$$(A \ominus B)^c = A^c \oplus \hat{B}$$

Morphological Internal Boundary Extraction:

Subtracting the eroded set $A \ominus B$ from the original set A isolates the exact boundary outline $\beta(A)$ of all chart components:

$$\beta(A) = A - (A \ominus B)$$

Observation: While gradient operators (Sobel/Prewitt) produce gray-level gradient magnitudes, morphological boundary extraction produces a closed binary contour with a precise single-pixel width (when using a 3×3 SE). This extracts the perimeter of chart bars and outlines alphanumeric characters without gradient thresholding.

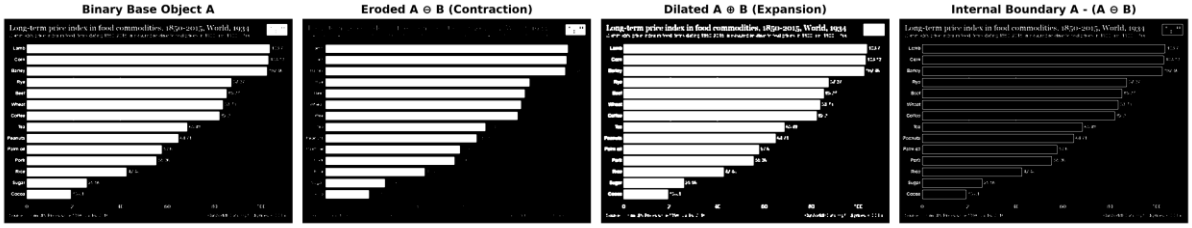


Figure 3: Morphological comparison and boundary extraction on chart_01.png: Original Binary Object A (far-left), Contraction via Erosion $A \ominus B$ (center-left), Expansion via Dilation $A \oplus B$ (center-right) and Morphological Internal Boundary $\beta(A) = A - (A \ominus B)$ (far-right).

6. Task 8.5: Iterative Morphology Experimentation ($k = 1, 2, 3$)

Applying morphological operators iteratively ($A \ominus kB = \big((A \ominus B) \ldots \ominus B\big)$) demonstrates progressive geometric degradation:

Operation	Iterations (k)	Foreground Pixels	Net Area Shift (%)	Structural Evolution & Topological Impact
Erosion	$k = 1$	165,962	-11.40%	Strips isolated single-pixel noise; bar outlines recede by 1 pixel.
Erosion	$k = 2$	150,828	-19.48%	Fine decimal points and thin characters ('1', 'I', '-') completely vanish.
Erosion	$k = 3$	136,468	**-27.14%**	Alphanumeric labels are fully eliminated; thin horizontal data bars fragment.
Dilation	$k = 1$	213,771	+14.13%	Bridges broken font characters; fills internal pinhole voids inside bars.
Dilation	$k = 2$	240,257	+28.27%	Letters fuse into solid character blocks; axis numbers merge with tick marks.
Dilation	$k = 3$	265,305	**+41.64%**	Adjacent parallel data bars fuse together, destroying inter-bar margins.

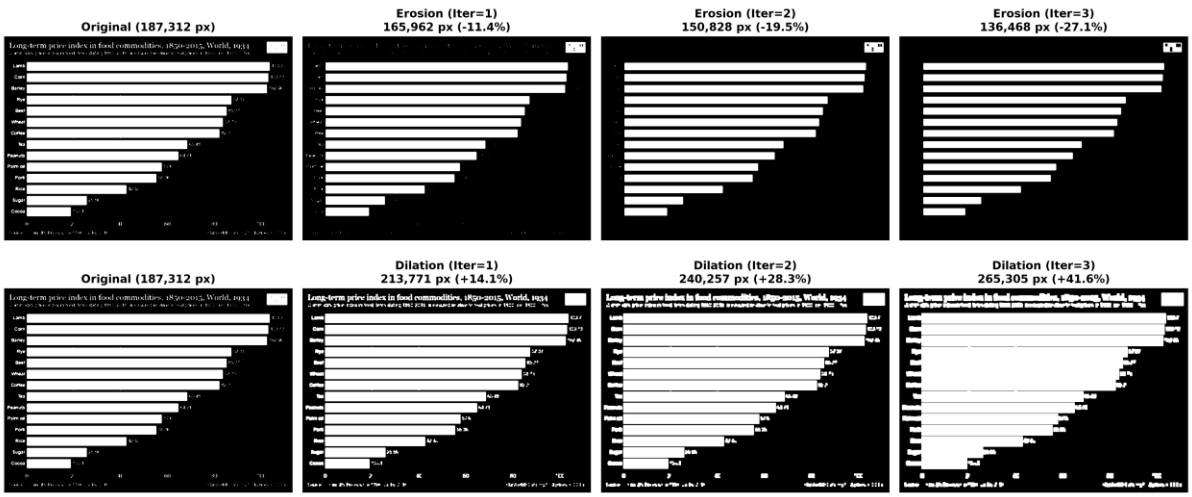


Figure 4: Iterative morphological progression using a 3×3 *Square SE: Top row illustrates progressive Erosion (*$k = 1,2,3$*); Bottom row illustrates progressive Dilation (*$k = 1,2,3$*).*

7. Application to Vision-Language Models & Visual Token Pruning

In LLaVA-1.5, the CLIP ViT-L/14 backbone divides input charts into non-overlapping 14×14 pixel patches, generating 576 visual tokens. Binary morphological processing serves two key roles in our pruning architecture:

1. **Noise Elimination via Erosion (**$k = 1$**):** Removes isolated stray pixels caused by JPEG compression ringing or scanner dust. This prevents empty background canvas patches from being falsely flagged as salient data tokens.
2. **Patch Densification via Dilation (**$k = 1$ **with Cross SE):** Text labels and thin line curves often contain single-pixel strokes that cross patch boundaries. Applying a single dilation pass bridges broken strokes into continuous foreground masks, ensuring that patches containing critical annotations register high saliency scores and are protected from pruning.

8. Challenges Faced & Solutions Implemented

1. **Foreground Polarity Inversion:**
    - *Problem:* Standard OpenCV morphological functions treat pixel value 255 as the foreground object. In raw charts, the background canvas is white (255) and the data bars are dark, which caused erosion to expand the background and shrink the chart bars.
    - *Solution:* Inverted the thresholding stage using cv2.THRESH_BINARY_INV, ensuring data structures are assigned 255 (foreground) and whitespace is assigned 0 (background).
2. **Text Disintegration under Large Structuring Elements:**
    - *Problem:* Using a 5×5 square SE eroded thin numerical labels completely on the first pass (19.48% loss).

- *Solution:* Standardized on the 3×3 **Cross-shaped structuring element** for document text regions, as its 4-connected shape preserves diagonal strokes while stripping orthogonal noise.

3. **Loss of Inter-Bar Gaps under Multi-Iteration Dilation:**
    - *Problem:* At $k = 3$ iterations, dilation expanded the foreground by $+41.64\%$, bridging adjacent bars into a single connected component and preventing individual bar counting.
    - *Solution:* Constrained morphological dilation to $k = 1$ iteration when generating foreground masks for token retention.

9. Learning Outcomes

- Implemented binary erosion and dilation from foundational set-theoretic definitions.
- Quantified morphological sensitivity across 3×3 square, cross-shaped and 5×5 structuring elements.
- Demonstrated morphological boundary extraction through set difference: $\beta(A) = A - (A \ominus B)$.
- Analyzed structural breakdown under iterative processing ($k = 1,2,3$), confirming that single-pass dilation bridges broken font strokes while single-pass erosion cleans background canvas noise.